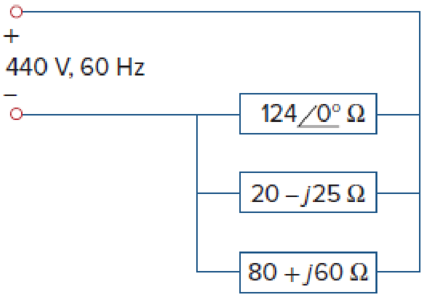
Chapter 11, Problem 38P

Problem

For the power system in Fig. 11.67, find: (a) the average power, (b) the reactive power, (c) the power factor. Note that 440 V is an rms value.

Figure 11.67:



Step-by-step solution

Step 1 of 5

Refer to Figure 11.67 in the text book.

Calculate the equivalent impedance Z_{eq} of the parallel connected impedance.

$$
\begin{aligned}
Z_{eq} &= 124\angle 0° \,||\, (20 - j25) \,||\, (80 + j60) \\
&= \frac{(124)(20 - j25)(80 + j60)}{(124)(20 - j25) + (80 + j60)(124) + (20 - j25)(80 + j60)} \\
&= \frac{384400 - j99200}{2480 - j3100 + 3100 - j800 + 9920 + j7440} \\
&= \frac{384400 - j99200}{15500 + j3540}
\end{aligned}
$$

$$
\begin{aligned}
Z_{eq} &= 22.181 - j11.466 \\
&= 24.969\angle -27.336°\ \Omega
\end{aligned}
$$

Therefore, the value of the equivalent impedance Z_{eq} is $24.969\angle -27.336°\ \Omega$.

Step 2 of 5

Calculate the value of the current I_{rms} flowing in the circuit.

$$I_{\text{rms}} = \frac{440}{Z_{eq}}$$

Substitute $24.969\angle -27.336°\ \Omega$ for Z_{eq}.

$$
\begin{aligned}
I_{\text{rms}} &= \frac{440}{24.969\angle -27.336°} \\
&= 17.622\angle 27.336°\ \text{A rms}
\end{aligned}
$$

Therefore, the value of the current I_{rms} flowing in the circuit is $17.622\angle 27.336°\ \text{A rms}$.

Step 3 of 5

Calculate the value of the average power P.

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Substitute $17.622\ \text{A rms}$ for I_{rms}, $440\ \text{V rms}$ for V_{rms}, $0°$ for θ_v and $27.336°$ for θ_i.

$$\begin{aligned} P &= (440)(17.622)\cos(0-27.336) \\ &= (7753.68)\cos(-27.336) \\ &= (7753.68)(0.8883) \\ &= 6.89\ \text{kW} \end{aligned}$$

Therefore, the value of the average power P is, $\boxed{6.89\ \text{kW}}$.

Step 4 of 5

Calculate the value of the reactive power Q.

$$Q = V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

Substitute $17.622\ \text{A rms}$ for I_{rms}, $440\ \text{V rms}$ for V_{rms}, $0°$ for θ_v and $27.336°$ for θ_i.

$$\begin{aligned} Q &= (440)(17.622)\sin(0-27.336) \\ &= (7753.68)\sin(-27.336) \\ &= (7753.68)(-0.4592) \\ &= -3.56\ \text{kVAR} \end{aligned}$$

Therefore, the value of the reactive power Q is, $\boxed{-3.56\ \text{kVAR}}$.

Step 5 of 5

Calculate the value of the power factor pf.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute $0°$ for θ_v and $27.336°$ for θ_i.

$$\begin{aligned} \text{pf} &= \cos(0-27.336) \\ &= \cos(-27.336) \\ &= 0.8883\ \text{leading} \end{aligned}$$

Power factor is leading since reactive power is negative.

Therefore, the value of the power factor pf is, $\boxed{0.8883}$.